

Particles on a straight line and plane

X20 (2020) — English translation

There is a straight line l and a plane a . They are perpendicular. Particle 1 can move in a straight line, and particle 2 can move along a plane; the mass of each particle is m . Particles are attracted according to the law of gravity. There is no friction.

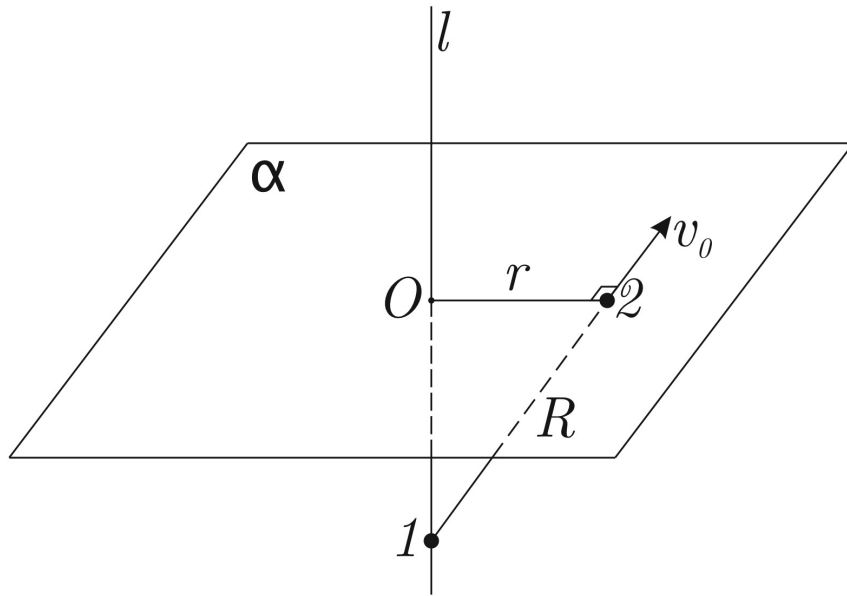


Figure 1: Figure 1.

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| A1 | Initially, particle 1 is at rest, and the second is set in motion. Its speed is perpendicular to the line connecting the particles, and the distance between the particles is R . Find the minimum value of the speed of the second particle v_{02} , at which the particles will be at an infinite distance from each other. | 3.00 |
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Now the particles are at equal distances from the point of intersection of the line and the plane, the distance between them is R . The particles are given equal velocities v_0 , lying in the same plane perpendicular to the plane α .

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| A2 | Find the value v_0 at which particle 2, after a long time, will finally stop at a finite distance from the origin and the value of the speed v_1 of the first particle in this case. | 3.00 |
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Source: <https://pho.rs/p/15>